

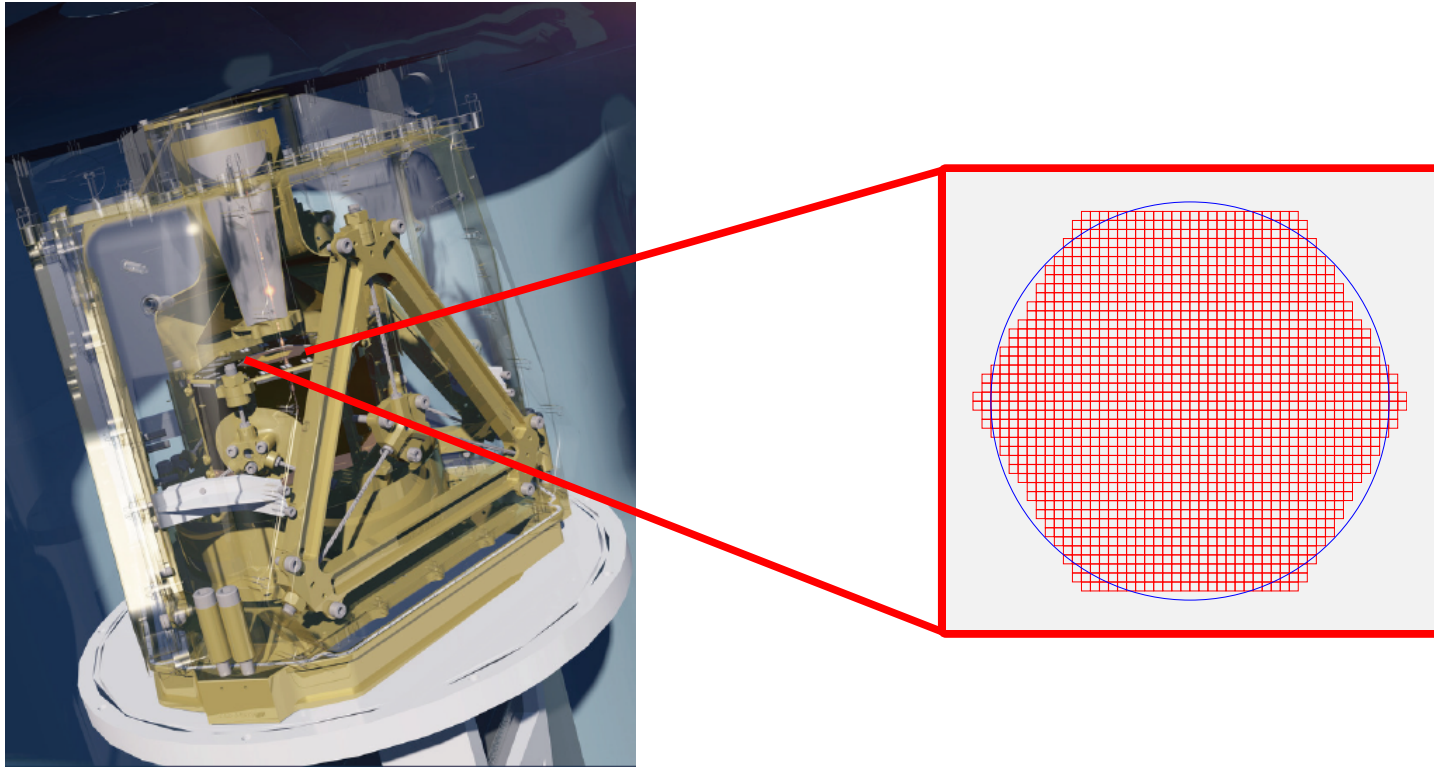
SIXTE Implementation of the *NewAthena X-IFU*

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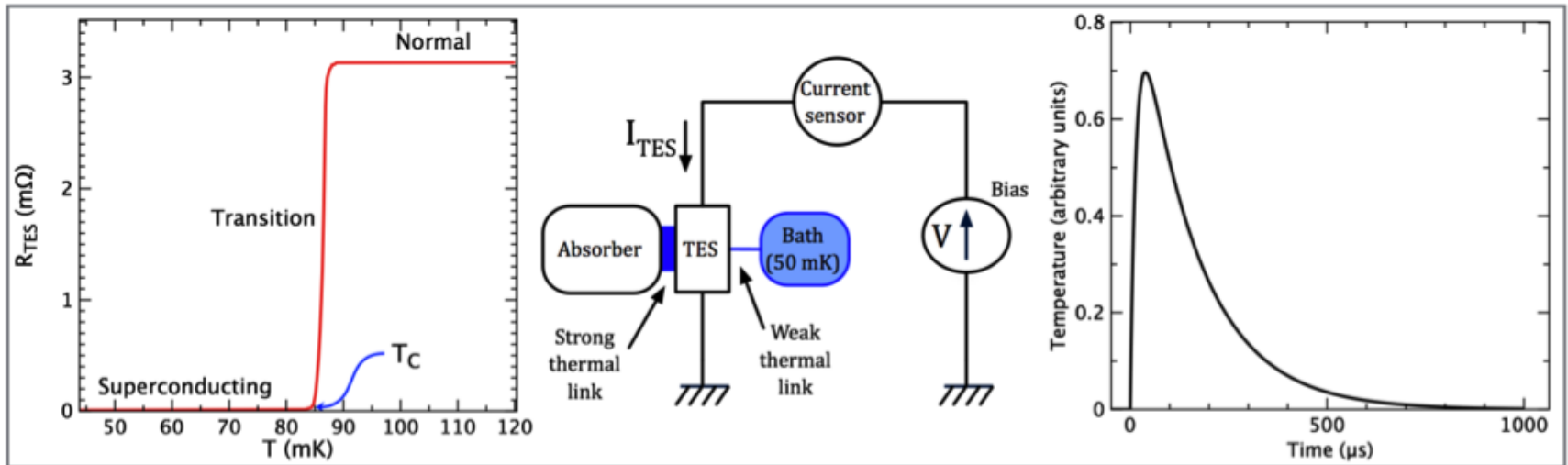
The X-ray Integral Field Unit (X-IFU)

- very high spectral resolution imaging (4 eV FWHM and a 4' FoV)
- 1504 TES (Transition Edge Sensor) pixels



The X-ray Integral Field Unit (X-IFU)

Pixels are single *Transition Edge Sensors*, operated at 50 mK
⇒ **measure temperature increase** of photon hitting the pixel



X-IFU Implementation in the end-to-end simulator SIXTE

sixtesim:

- full detector array
- full imaging implemented
- fast detection simulation using response matrices + microcalorimeter-specific effects

⇒ **science simulations**

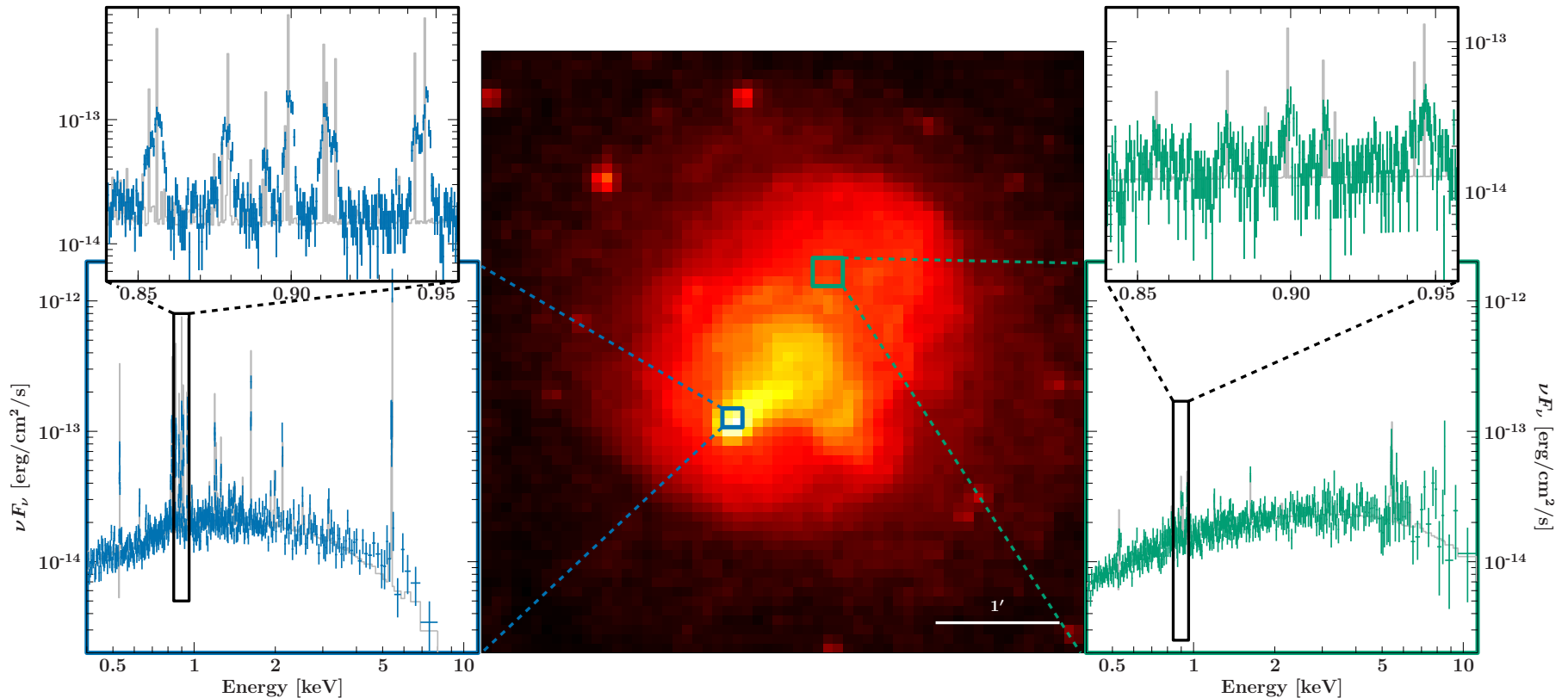
tessim/xifusim + SIRENA

- Simulation of TES physics and pulse reconstruction
- Slower than `sixtesim`, but much better physics
- pixel interaction (crosstalk)

⇒ **Input for** `sixtesim`

⇒ **physics-based** `tessim/xifusim` **results converted to be used in the fast and general** `sixtesim` **simulation (event grading, crosstalk, ...)**

Example: SIXTE X-IFU simulation of a Galaxy Cluster

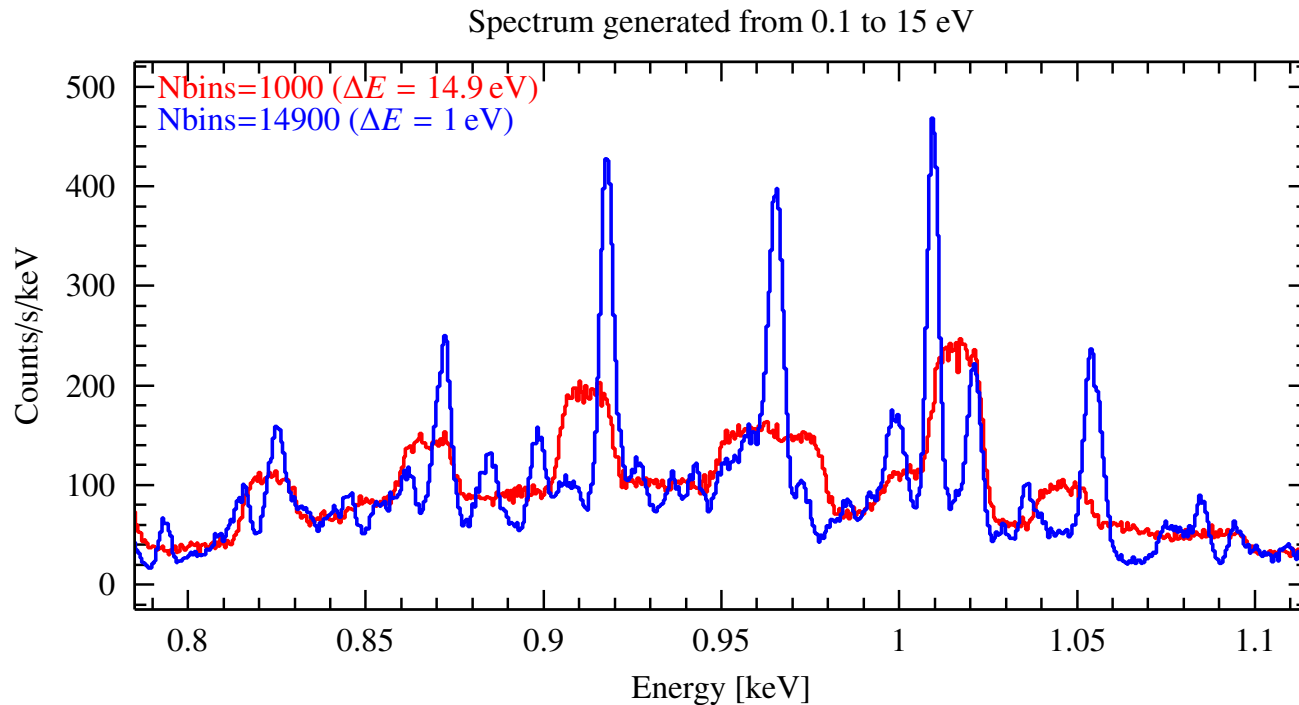


SIXTE describing, simulating, and analyzing complicated sources
X-IFU spatially resolved high-resolution spectroscopy

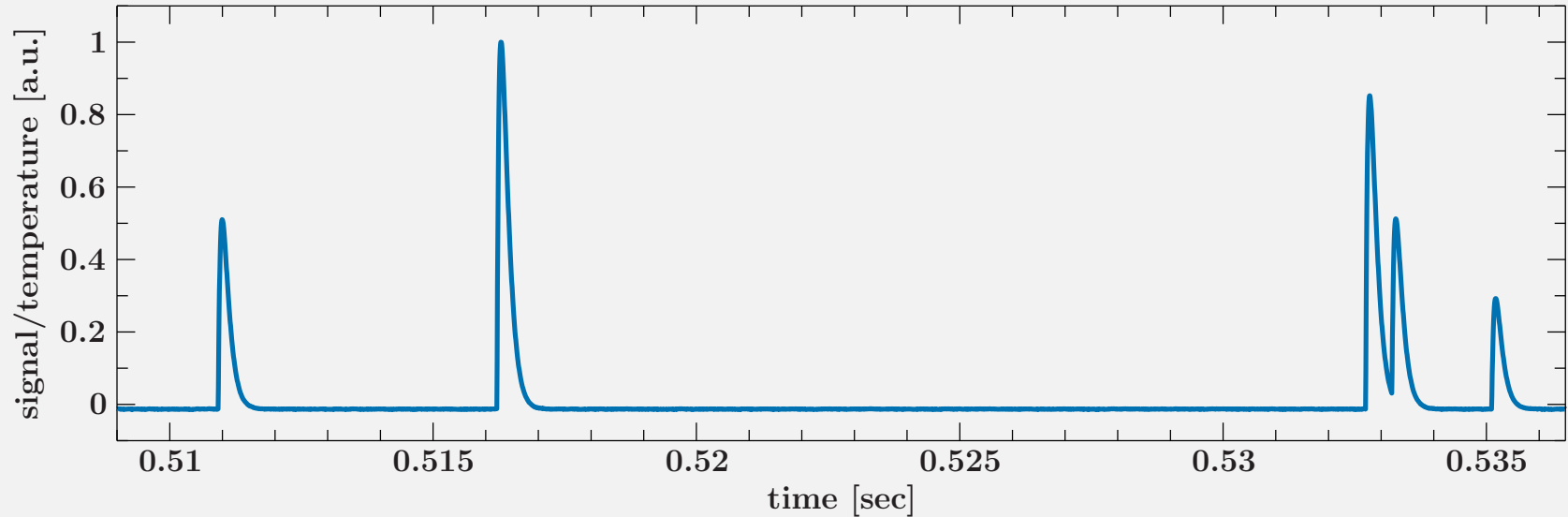
Simput Binning

One consideration for any high-resolution instrument: **binning your SIMPUT!**
When calling `simputfile` with an XSPEC or ISIS file, the spectrum is evaluated **once**, on the energy grid defined by the parameters `Elow`, `Eup` and `Nbins`.

Setting this value too low will have strange effects on a spectrum with high-resolution features (e.g. an APEC):



Grading

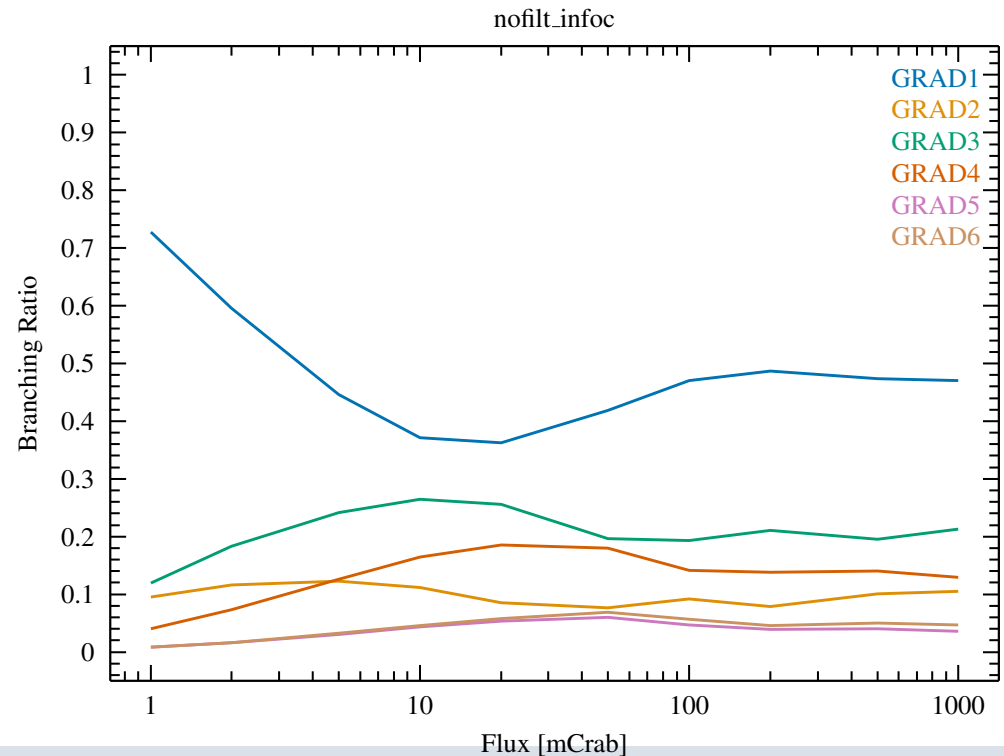


pulses with **smaller separation yield lower energy resolution**
⇒ **Event Grading** depending on the source flux

Grading

Grade	Δt since previous pulse	Δt until next pulse	Energy res.
(1) Very High res.	≥ 24.0 ms	≥ 55.2 ms	4 eV
(2) High res.	≥ 24.0 ms	≥ 26.8 ms	~ 4 eV
(3) Intermediate res.	≥ 12.0 ms	≥ 11.1 ms	4.2 eV
(4) Medium res.	≥ 12.0 ms	≥ 3.16 ms	5 eV
(5) Limited res.	≥ 12.0 ms	≥ 1.20 ms	7 eV
(6) Low res.	≥ 12.0 ms	≥ 0.05 ms	30 eV

Throughput per grade
 (“branching ratio”) strongly
 depends on source flux!



Crosstalk

unintended transmission of information between signal channels

Different types of crosstalk:

- **thermal** coupling of two pixels (physical neighbors)
- **electrical** coupling due to e.g. mutual inductance
- coupling due to **multiplexed** readout (TDM)

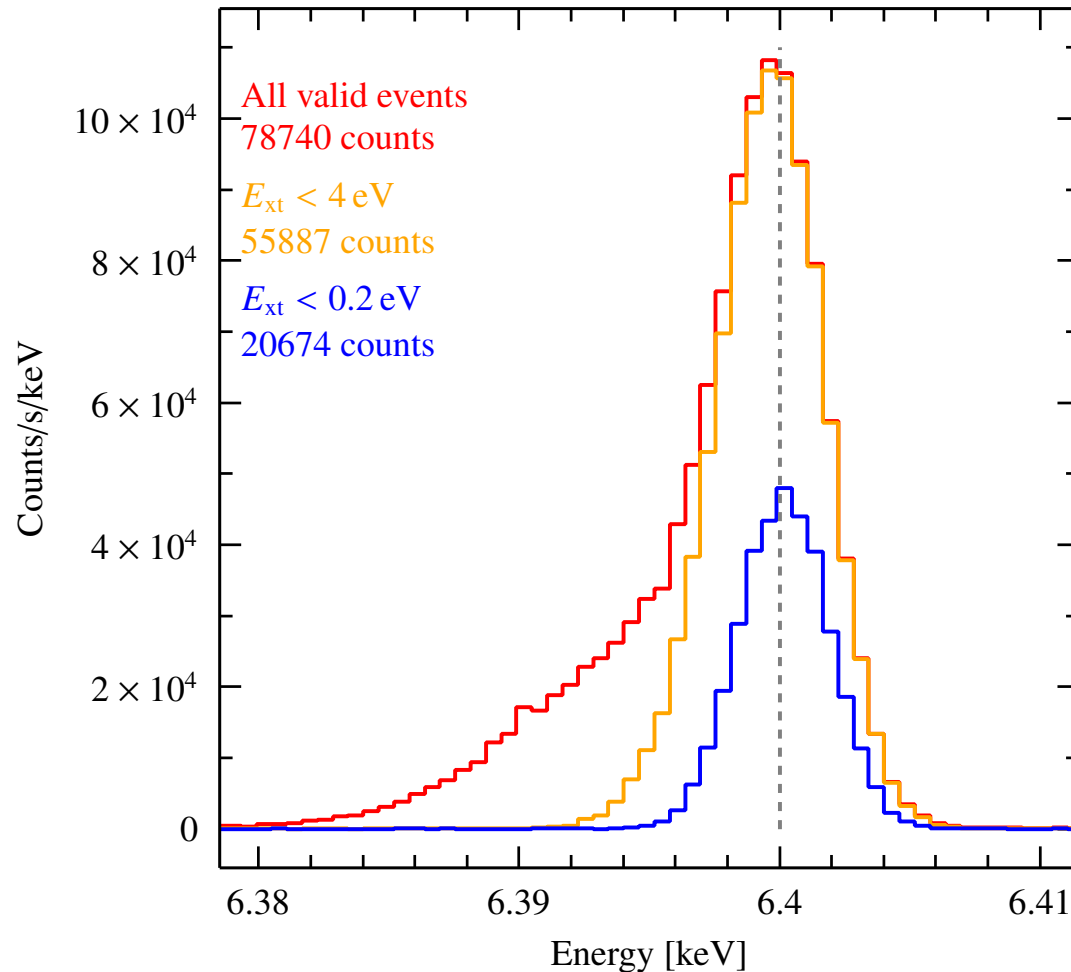
→ implemented in SIXTE

crosstalk effect on events is predictable

Crosstalk

Simulation of a narrow emission line (1 Crab)

Total Nr. of photons: 222254 (2222.54 cts/s)

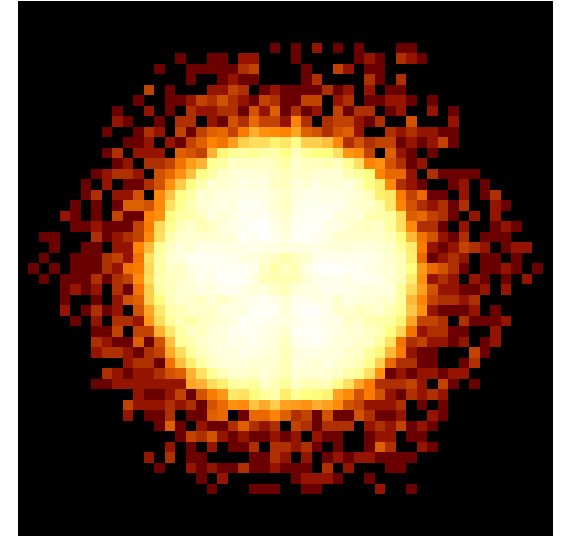
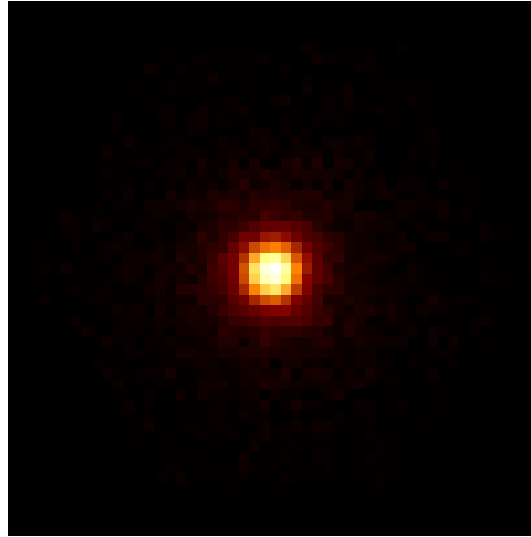


Crosstalk effect is **predictable**,
can remove events strongly
affected by crosstalk.

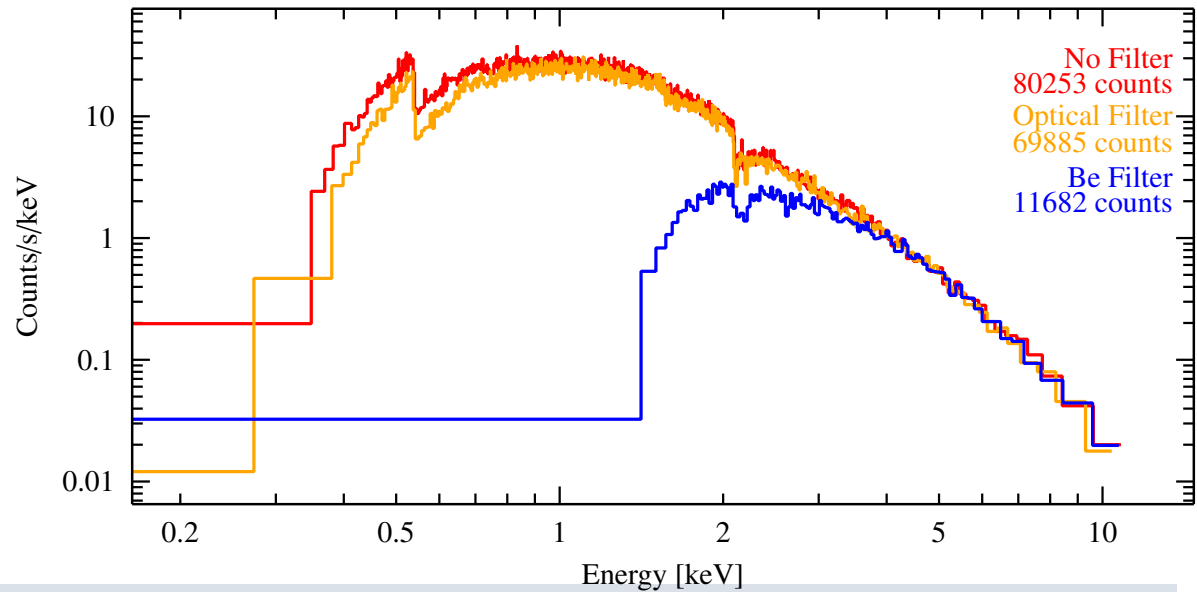
trade-off between
energy resolution and
throughput

Bright source mitigation

Defocusing distributes flux across detector.

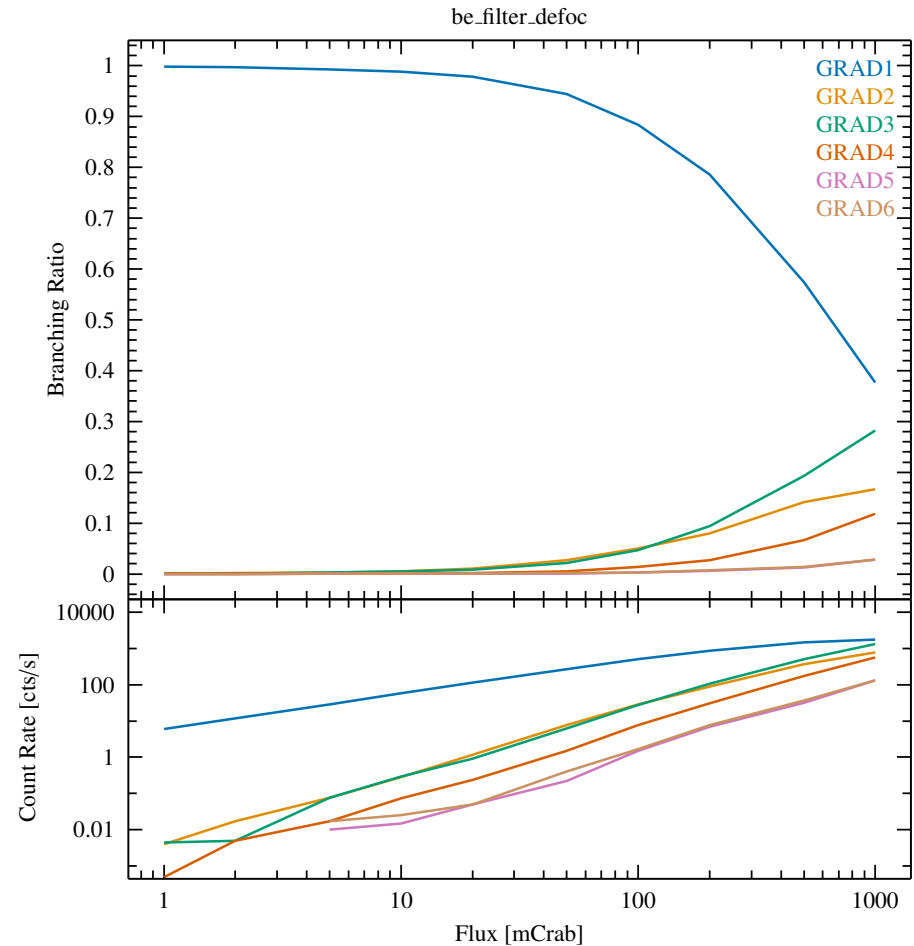
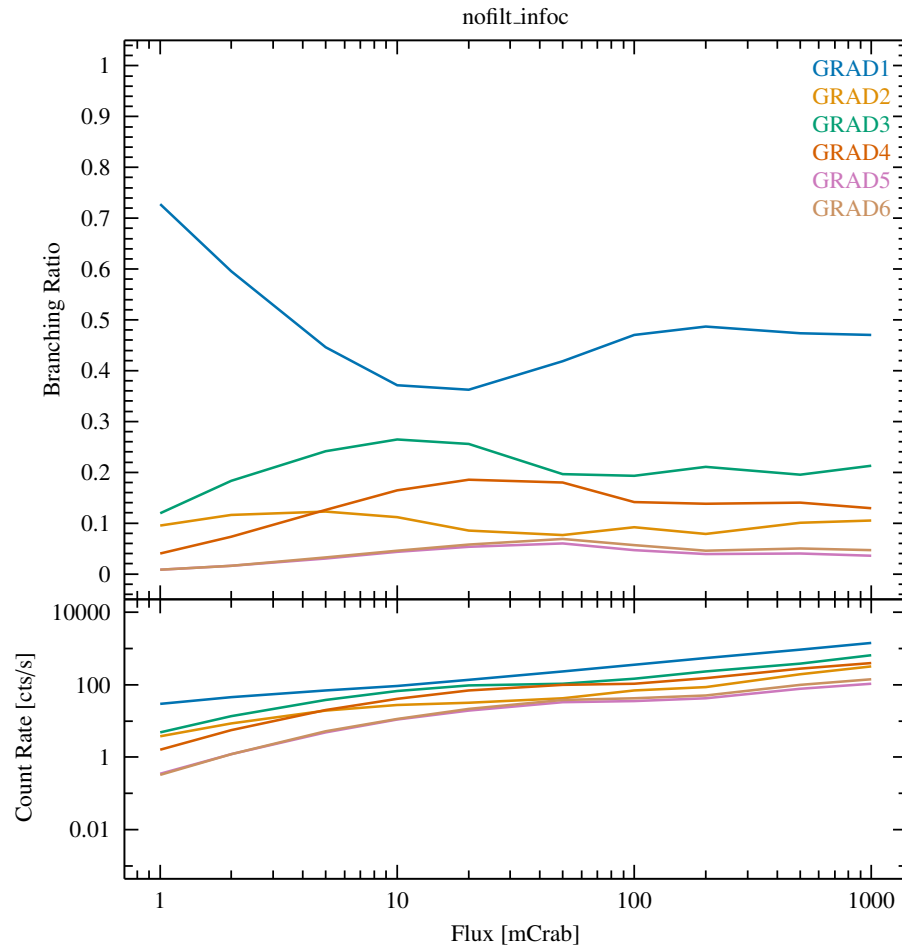


Filters reject flux at low energies



Bright source mitigation

Use **defocusing + filter** to improve **high-resolution throughput**

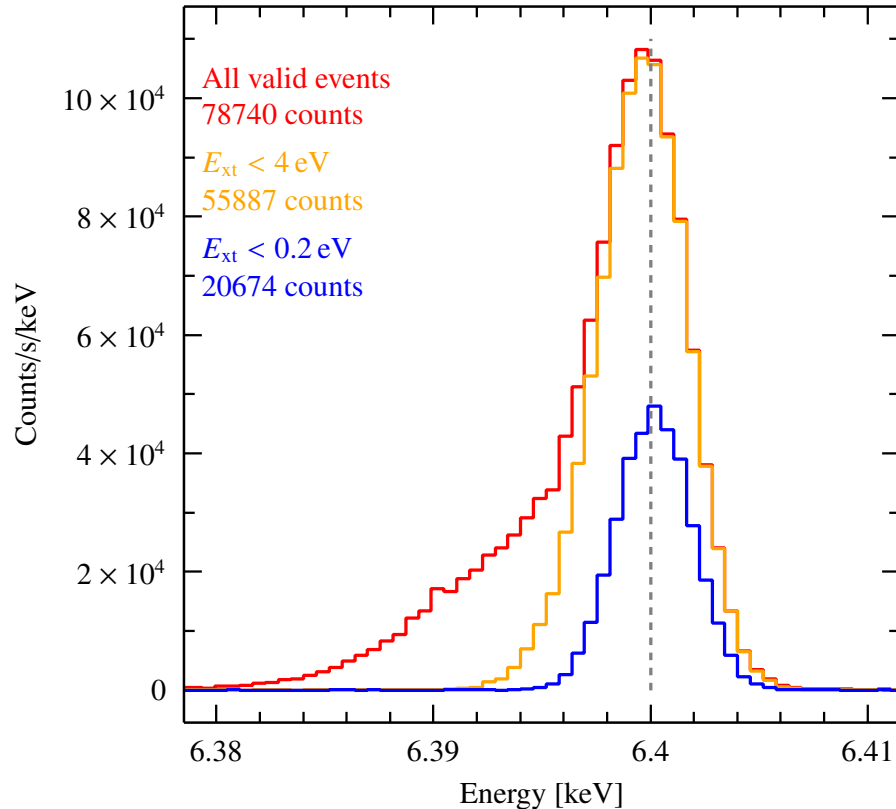


Bright source mitigation

Use **defocusing** to improve **number of valid events** and **crosstalk performance**

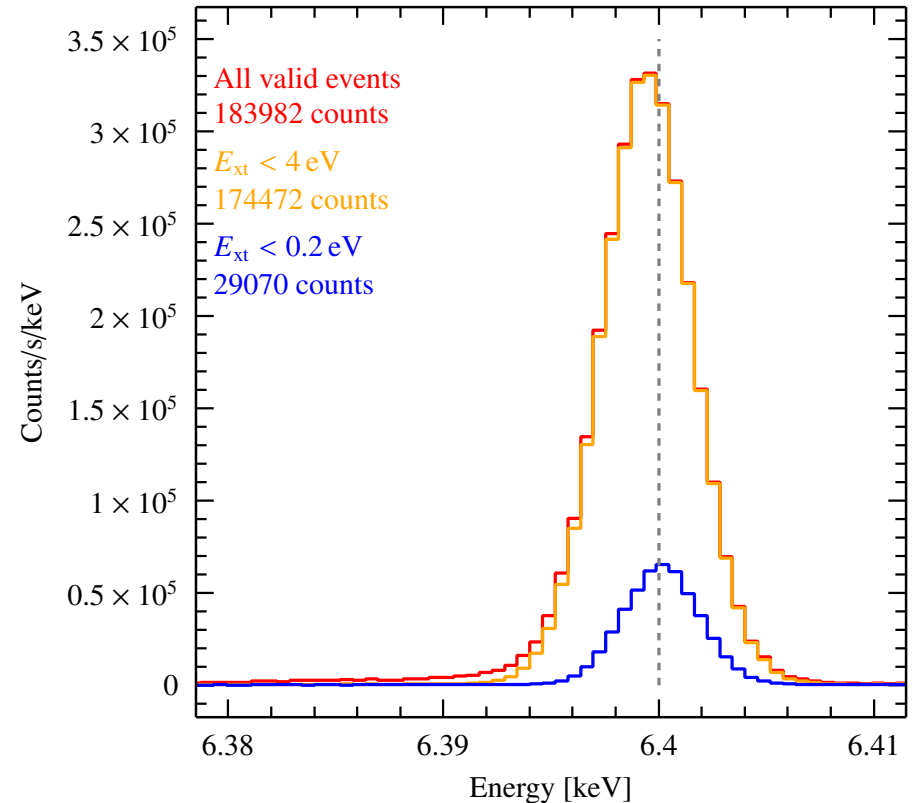
In-Focus

Total Nr. of photons: 222254 (2222.54 cts/s)



Defocused

Total Nr. of photons: 222254 (2222.54 cts/s)



Bright source mitigation

X-IFU event files contain additional information to **characterize bright source effects**:

- **Extension columns** giving the event grade (GRADING), crosstalk energy shift (E_XT) and number of events causing crosstalk (N_XT)
 - These can be used for **filtering**!
- **Header keywords** for number of valid/invalid events and overall grade distributions

```
NEVT      =      222791 / number of events simulated
NVALID    =      160569 / number of events with valid grades
NINVALID  =      62222 / number of events with invalid grades
NGRAD1    =      91267 / number of events with grade 1
NGRAD2    =      14152 / number of events with grade 2
NGRAD3    =      28479 / number of events with grade 3
NGRAD4    =      17220 / number of events with grade 4
NGRAD5    =       4602 / number of events with grade 5
NGRAD6    =       4849 / number of events with grade 6
```

■ GRADING 1I	■ N_XT 1I	■ E_XT 1D keV
Modify	Modify	Modify
4	1	-2.284780472218E-05
-1	0	0.000000000000E+00
-1	0	0.000000000000E+00
-1	0	0.000000000000E+00
-1	0	0.000000000000E+00
3	3	-7.209926662652E-05
4	1	-2.666993170494E-05
-1	0	0.000000000000E+00
3	0	0.000000000000E+00
1	11	-1.414753001797E-03
1	7	-1.828143950962E-04
1	2	-1.836882067134E-04
1	9	1.142133631182E-04
1	16	-1.324613853145E-03
1	4	-6.917426857100E-05
1	0	0.000000000000E+00
3	2	1.652283974995E-03
5	1	-6.102067764273E-04
-1	0	0.000000000000E+00
3	5	1.800124248924E-03

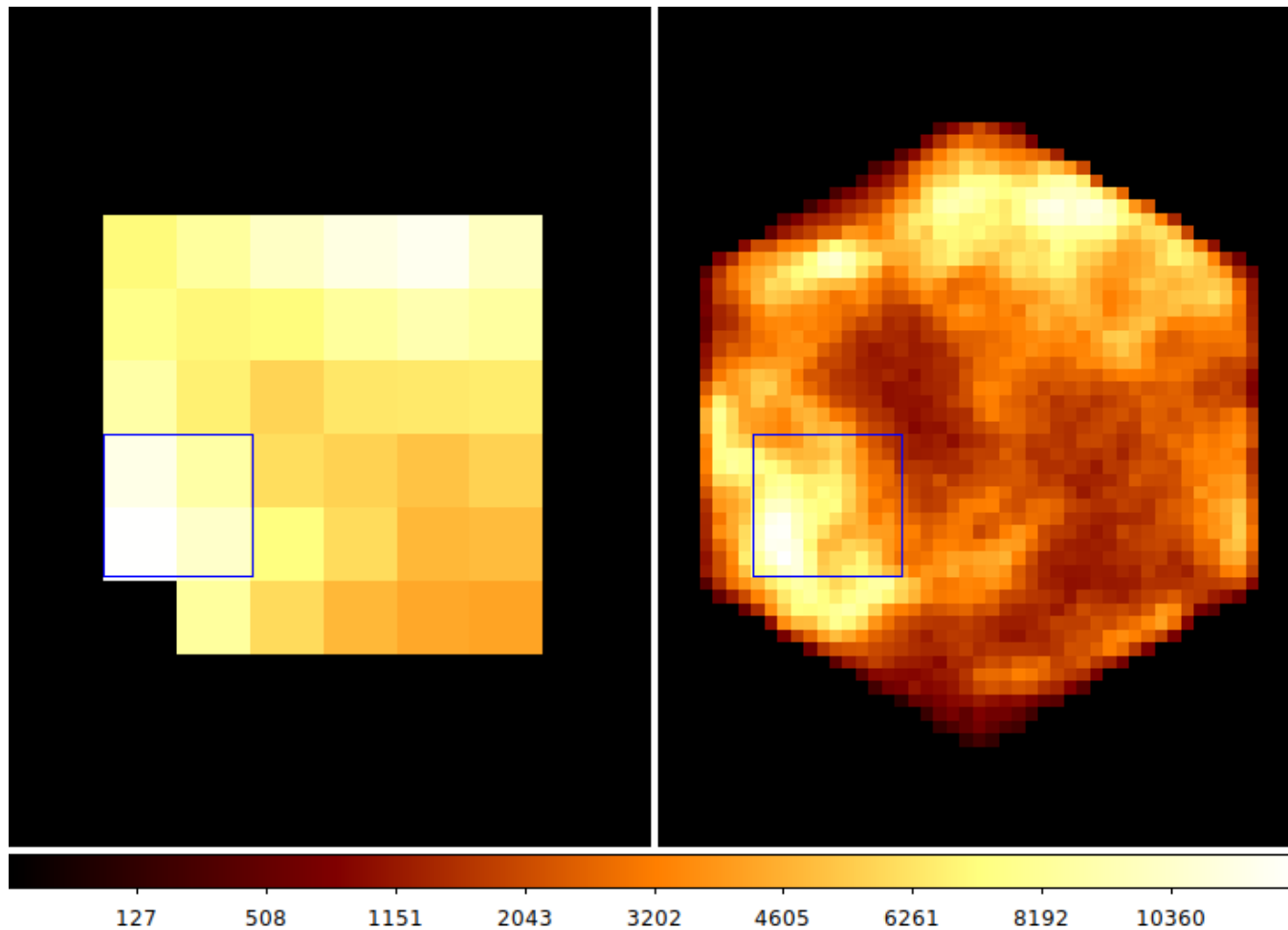
XRISM Resolve

How do we simulate XRISM Resolve with SIXTE?

Simply use `sixtesim` with a different XML!

This simulates the Grading effect as well. Crosstalk is not yet implemented, due to lack of data.

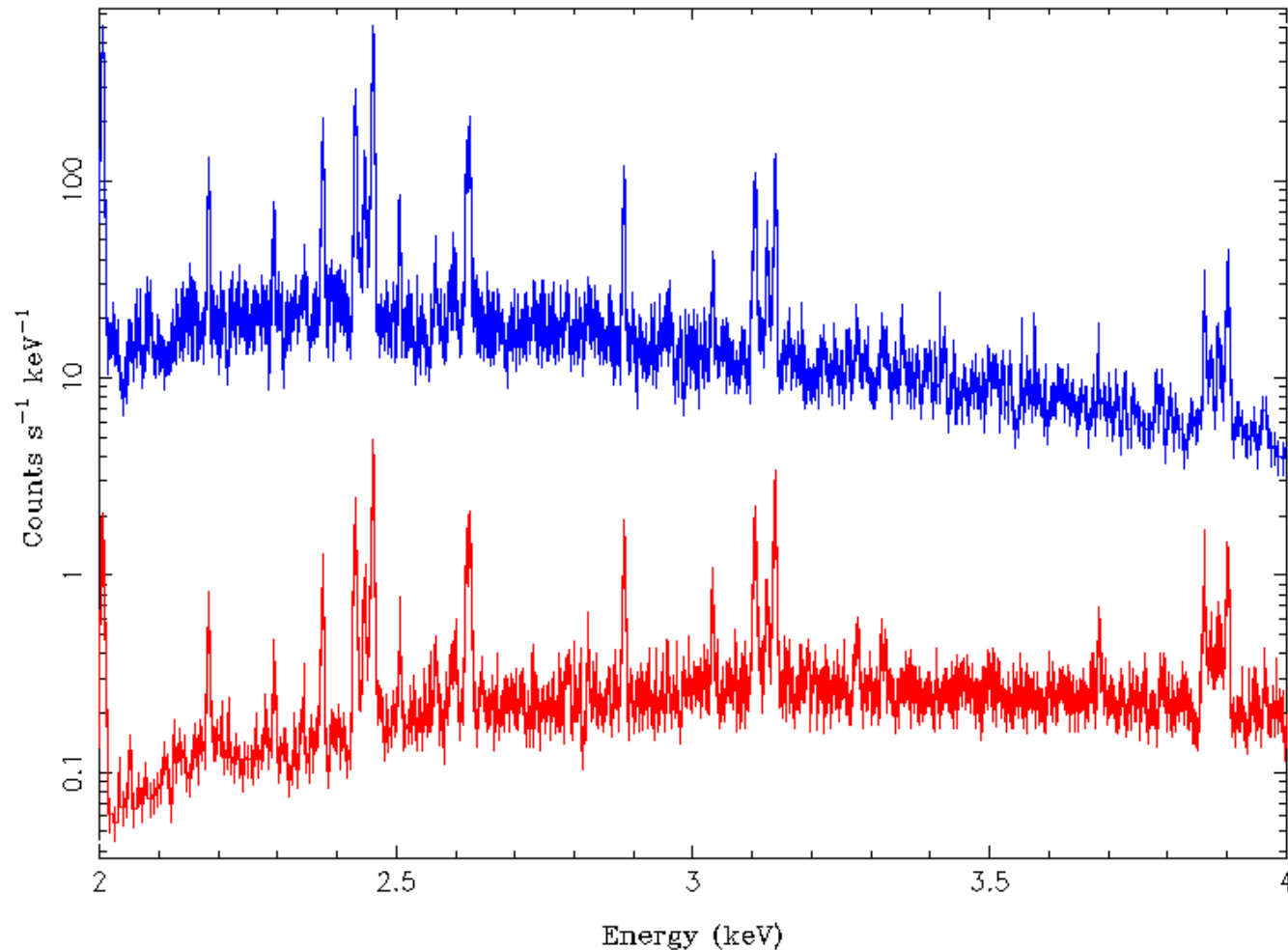
Example: Cas A, center



50 ks Resolve with Gate Valve

1 ks X-IFU with Be Filter

Example: Cas A, center



50 ks Resolve with Gate Valve

1 ks X-IFU with Be Filter

Summary: The X-IFU with SIXTE

- 1504 TES pixels in a hexagonal array
- 4' FoV
- higher flux ($> 10\text{mCrab}$) reduces energy resolution and throughput
- science simulations with `sixtesim`, taking the most important TES physics effects into account
- physics input to the simulation pipeline by `tessim/xifusim`